

الفصل الثاني
2024/2023

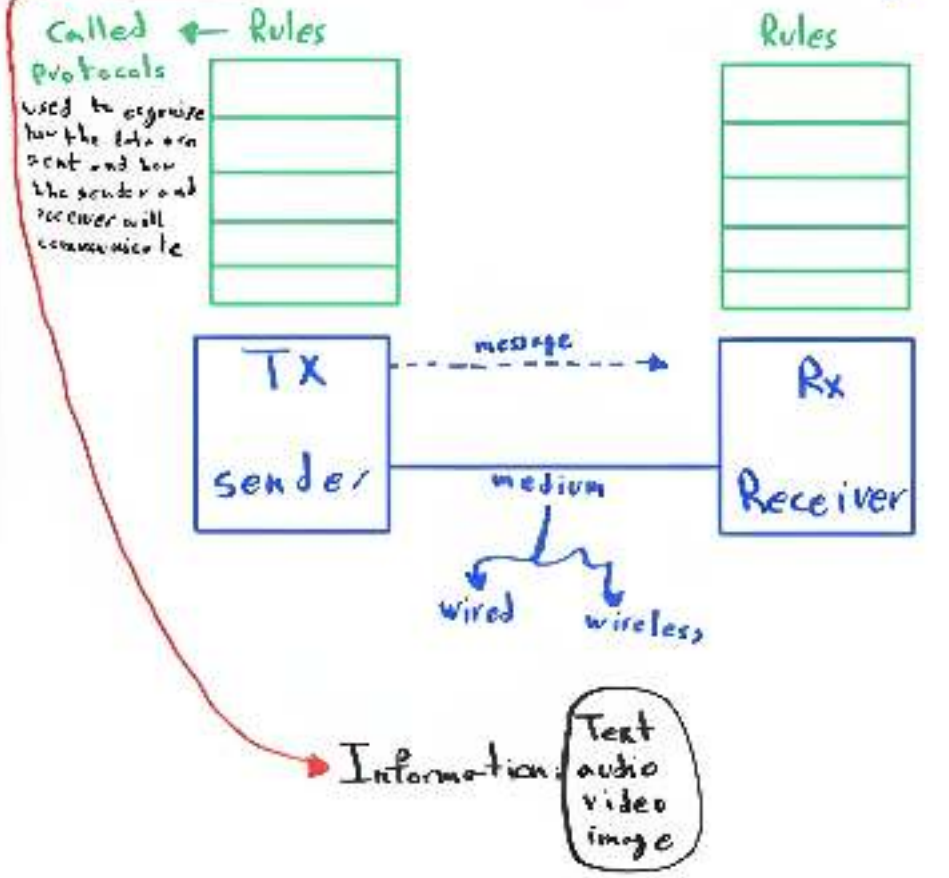
اتصالات وتراسل البيانات

د.أمجد الهندي

Team ZerOne
Computer & Networks Engineers

Data Comm. & Networking

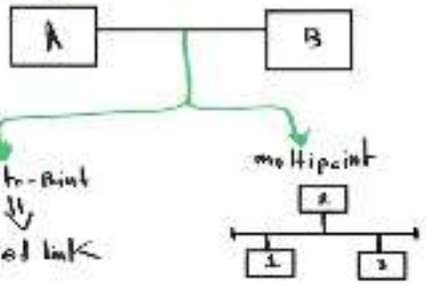
Lecture 1



Modes

- **Simplex mode** & **Broadcasting**
  - we can think of it like Satellite
  - the comm. process is from one side
  - Satellite → sender
  - Dishes → receiver
- **half-duplex mode**
  - we can think of it like the walkie-talkie
  - each of devices can transmit and receive but
  - Not Simultaneously
- **full-duplex mode**
  - we can think of it like phone calls
  - because both devices can transmit and receive data.
  - SimuHoneously

Network: Any set (two or more) of connected devices capable of comm.

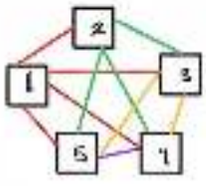


Could be called → host, end-system, node, terminal (like → large computer, desktop, laptop, cellular phone)

or could be called → Connecting device (like → Router → connects the network to other networks; switch → connects devices together; modem → change the form of data (modulator-demodulator))

Topologies: (more info in textbook page 45)

Mesh

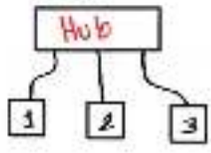


number of nodes: n = 5

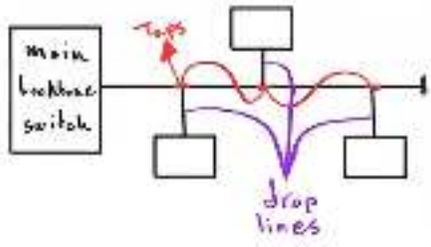
number of full duplex links: $\frac{n(n-1)}{2} = \frac{5(4)}{2} = 10$

number of I/O ports: (n - 1)

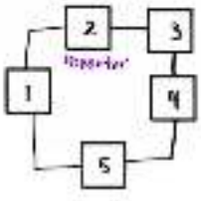
Star



Bus

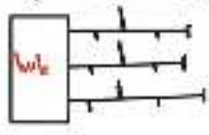


Ring



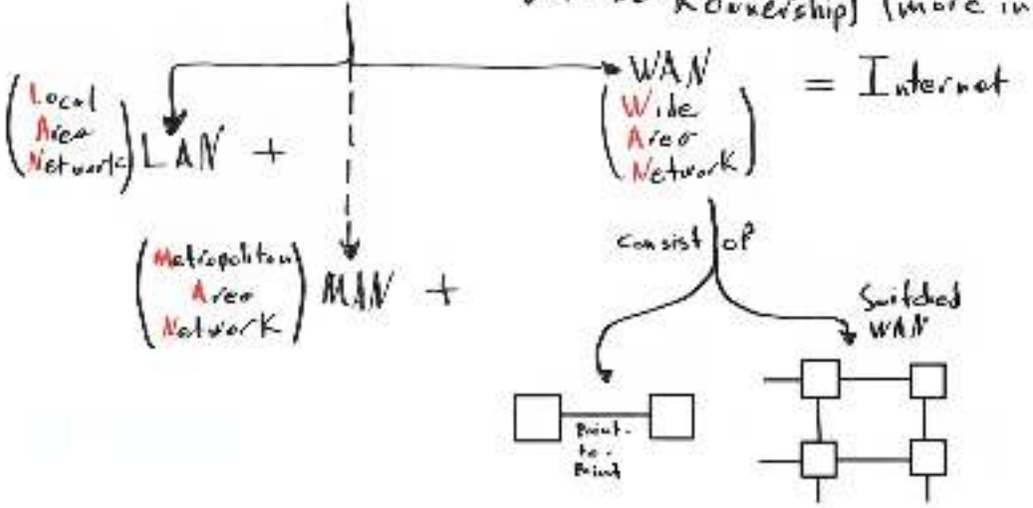
→ For knowledge only

Hybrid which is multiple topology method together



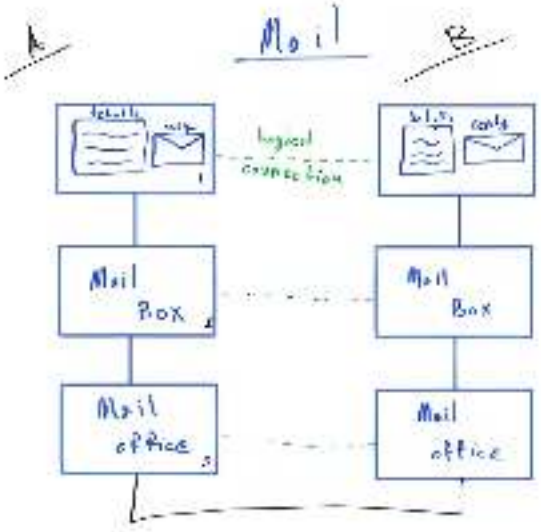
→ Star with bus

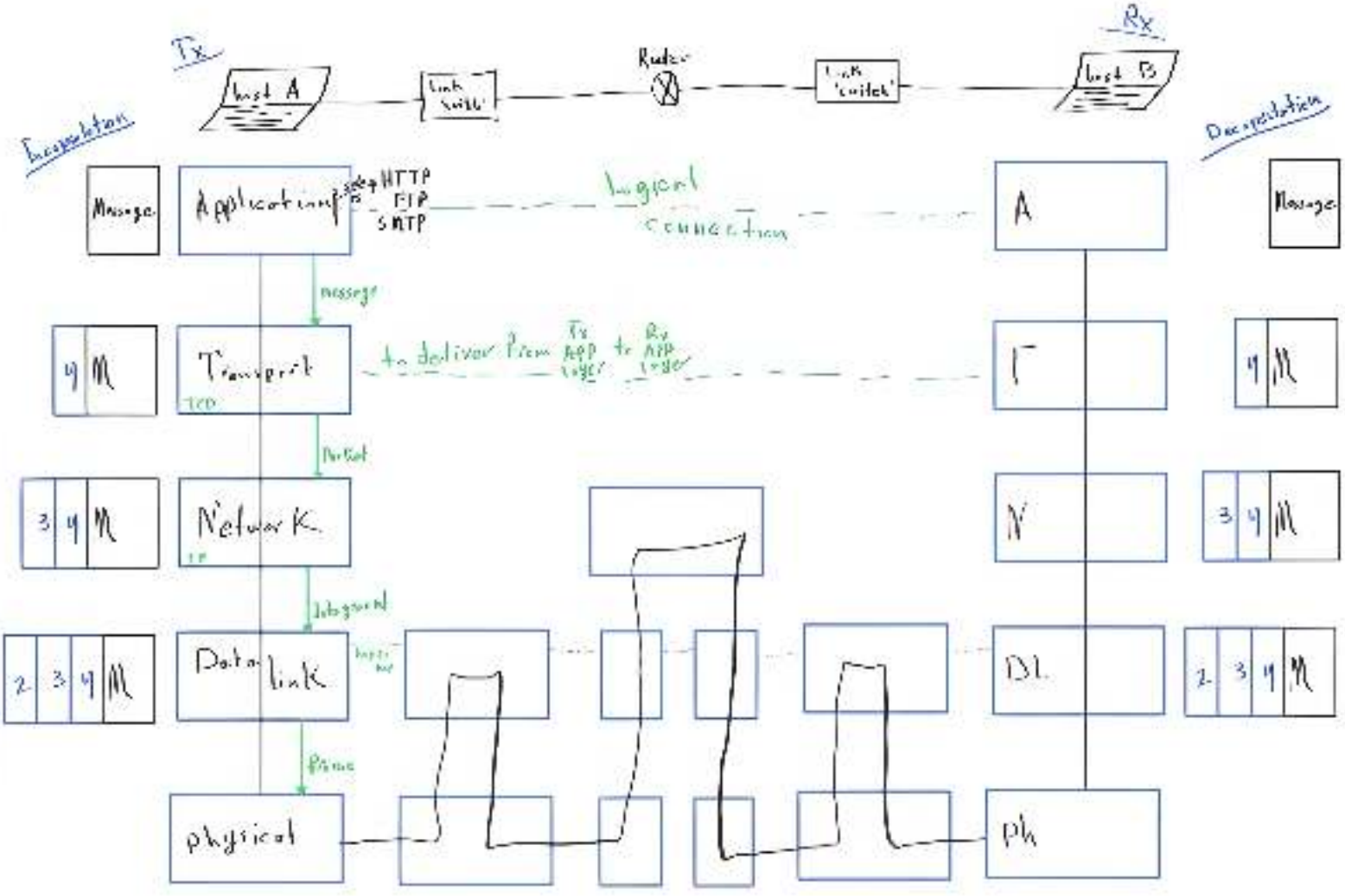
Network Types: (coverage, size & ownership) (more info in textbook page 52)



Lecture 3

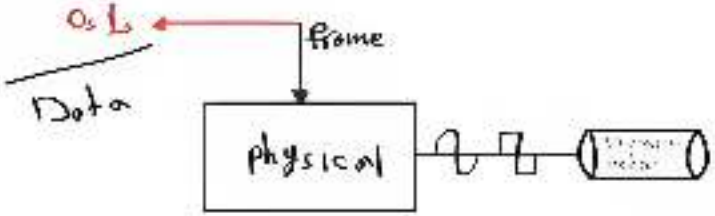
Network Models: TCP/IP

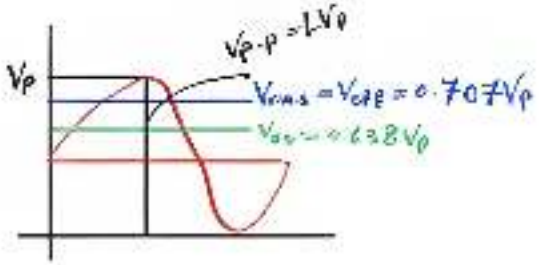




Lecture 4

Introduction to physical layer





Data Versus Signals

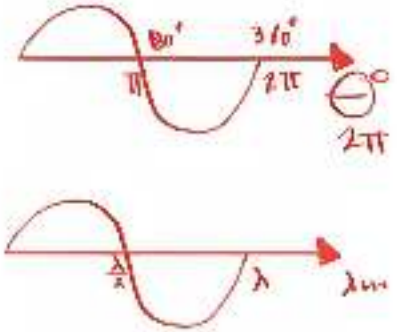
- Analog
- Digital

- Analog
- Digital

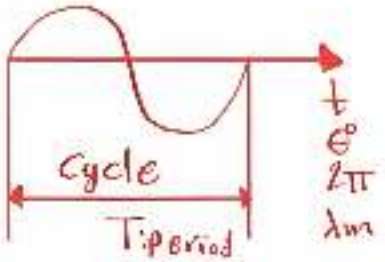
- Periodic: $x(t+T) = x(t), \ -\infty < t < \infty$
- Non-Periodic: $x(t+T) \neq x(t)$

$\omega = 2\pi f = \frac{2\pi}{T}$

Angular Velocity Rad/s



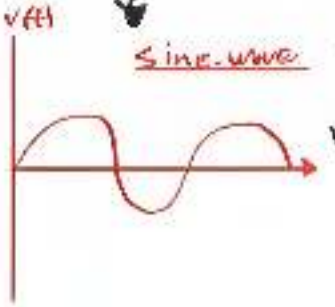
$\lambda = \frac{c}{f}$ → 3×10^8 m/s



$T = \frac{1}{f} \rightarrow f = \frac{1}{T}$ → period

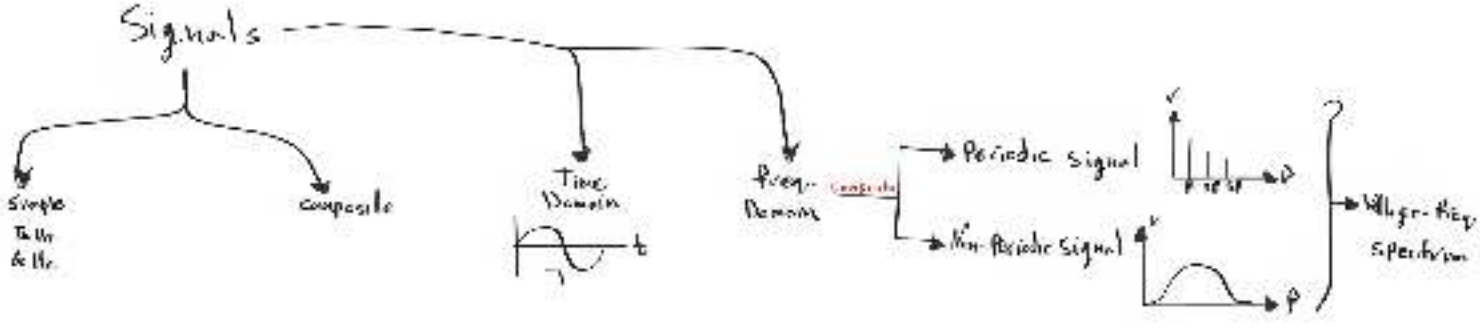
freq in Hz: KHz, MHz, GHz

period: ns, ms, us



Phase: Measure of relative position in time.

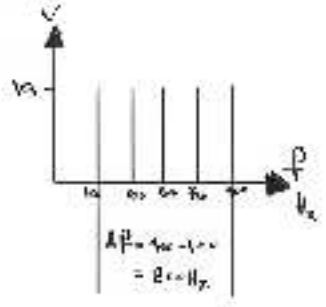
Lecture 5



$$\text{Bandwidth} = B = \Delta f = f_{max} - f_{min}$$

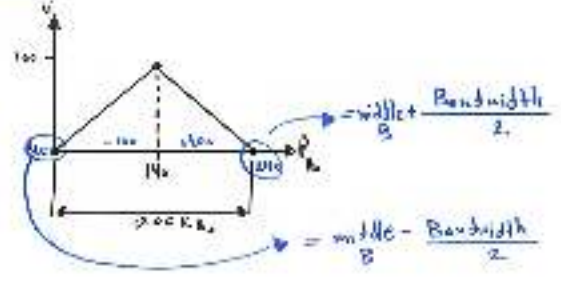
Ex: If a periodic signal is decomposed into 5 sine-waves with Freq: 100, 300, 500, 700 & 900 Hz.
What is the bandwidth?
Draw the spectrum assuming all components have V_{max} = 10V.

Sol:



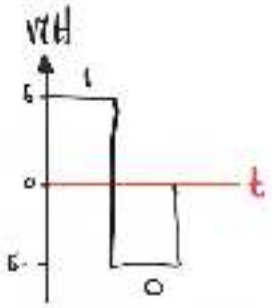
Ex: A nonperiodic composite signal has B = 200 KHz with middle freq of 140 KHz with amp. of 10V
Sketch the frequency with the amp. Draw the spectrum?

Sol:



Lecture 6

Digital Signals

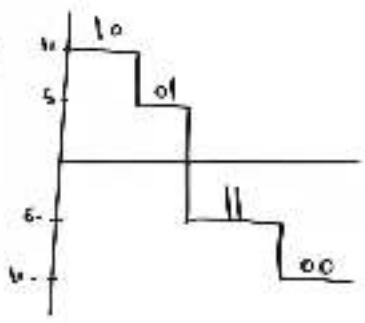


Number of bits/level = $\log_2 L$ ← L: Number of levels

one bit/level: $1 = \log_2 2$

2 bits/level: $2 = \log_2 4$

3 bits/level: $3 = \log_2 8$



Concept → BitRate
bps

Ex. Assume we need to download text document at the rate of 100 pages per second
what is the required bitrate of the channel?
if Page is: 24 line × 80 character/line
Consider 1 char = 8 bits

Sol:

24 × 80 × 100 = 192000 char/s

P X

1536,000 bit/s

= 1.5 Mbps

Concept → Bit length

Bit length = Propagation Speed * duration of bit

$m = \frac{m}{s} * s$

$\lambda = \frac{c}{f}$

$= cT$

Ex: what is the bit rate for high-definition TV (HDTV)?



1920 × 1080 Pixels

movement: 30 frames/s

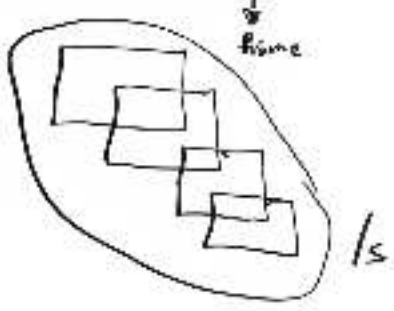
1 Pixel = 24 bit

Moving picture → frame

1920 × 1080 × 30 × 24

= 1,492,992,000 bps

= 1.492 Gbps

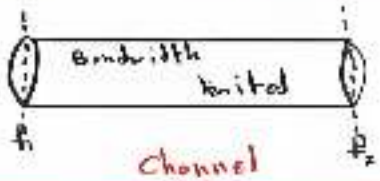


baseband Signal
orginal Signal
message Signal
modulating signal
Information signal

→ No change

Lecture 7

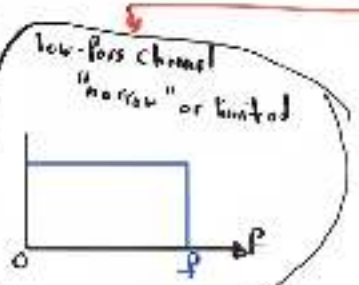
Transmission of digital Signals



BaseBand Transmission
"wired" short distances

Channel

low-Pass channel
"narrow" or limited



low-Pass channel
"wide"



Rough Approximation

Data rate: N

$f_1 = \frac{N}{2}$

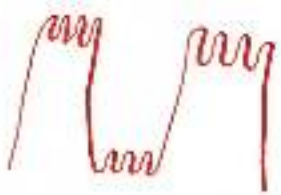


more harmonic

Better Approximation

$f_1 = \frac{5N}{2}$

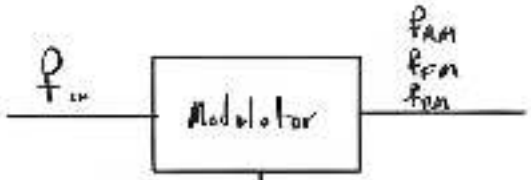




Broadband Transmission
"wireless" & long distance

→ Antenna ∝ Change

Modulation



$$\lambda = \frac{c}{f}$$

lecture 8

Transmission Impairments

- Attenuation & Amplification
- Distortion
- Noise
  - Thermal
  - Crosstalk
  - Impulse
  - Induce





Decibel: (db) logarithmic measure of ratio of signals levels

$$SNR = 10 \log_{10} \frac{P_{signal}}{P_{noise}}$$

$$Gain\text{"db"} = 10 \log_{10} \frac{P_{out}}{P_{in}}$$

Ratio	db	Ratio	db
1	0	0.5	-3.01
2	3.01	10^{-1}	-10
10^1	10	10^{-2}	-20
10^2	20	10^{-3}	-30
10^3	30		

$dbw \longrightarrow 10 \log_{10} \frac{P}{1w}$

$dbm \longrightarrow 10 \log_{10} \frac{P_{mw}}{1mw}$

$$V_{db} = 20 \log_{10} \frac{V_{op}}{V_{ip}}$$

Lecture 9

Data Rate Limits

(bps)

N_{max} (bps) — max data rate

Noiseless Channel ← By Nyquist

Bandwidth of the channel

Max Bit Rate = $2B \log_2 L$

L → Number of levels

Q: channel noiseless with $B = 3\,KH_z$ (Bandwidth)

transmitting two signal levels (# of levels)

Calculate max bit rate (N)?

Sol:

$N_{max} = 2(3\times10^3) \times \log_2 2$

$= 6000$ bps

Q: channel noiseless with $B = 20\,KH_z$ (Bandwidth)

and max bitrate (N) (265 K bps)

how many signal level (L) is it transmitting?

Sol:

$265K10^3 = 2 \times 20K10^3 \log_2 L$

$L = 98.7 \approx 98$

Channel is Noisy ← By Shannon

Bandwidth of the channel

Channel Capacity $= B \log_2(1+SNR)$

SNR → Signal to noise ratio

$SNR = \frac{P_s}{P_n}$

Q: Consider an extremely noisy channel (SNR ≈ 0)

Calculate channel capacity? ($B = 6$ MH$_z$)

Sol:

$Ch.cp. = 6M10^6 \times \log_2(1+0)$

$= 0$

Q: A telephone line of $B = 3KH_z$

& $SNR = 3162$. Calculate Ch.cp?

Sol:

$Ch.cp = 3K10^3 \times \log_2(1+3162)$

$= 34881.23$

≈ 34881

Lecture 10

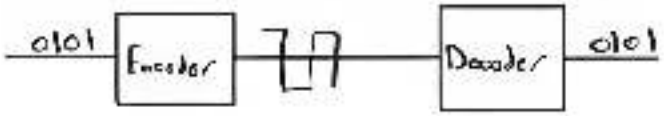
Performance

- **Bandwidth** → Hz, bps
- **Throughput:** bps
  - Practically
  - Data Rate / Theoretically
- **Latency** "delay" = Propagation time + Transmission time + Processing time + Queuing time (time)
  - Propagation time: $t = \frac{\text{Distance}}{\text{speed}}$
  - Transmission time: $t = \frac{\text{Message Size}}{\text{Bandwidth}}$ "L" / bps

Lecture 11

Digital Transmission

D/D conversion:
- line coding
- Block coding
- scrambling

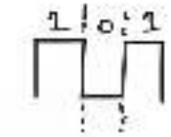


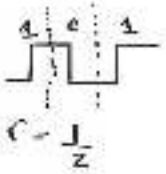
Basics:

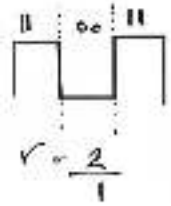
Data Element vs Signal Element

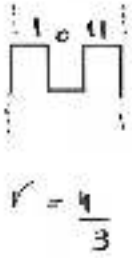


r: number of data elements carried by each signal element









N: bit Data Rate (bps) vs S: Signal Rate (Baud)

$S = \frac{N}{r}$

$S_{average} = C * \frac{N}{r}$

C = factor

H_z (bps) → Bandwidth

traffic?

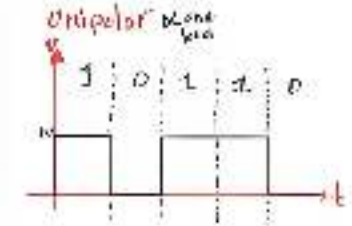
$B_{min} = C * \frac{N_{max}}{r}$

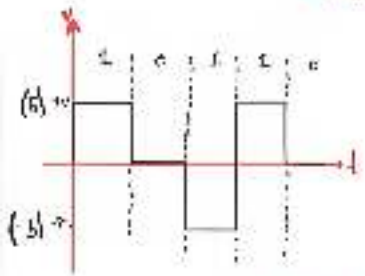
Advantages:
- ✓ Immune to Noise
- ✓ self-synchronization
- ✓ built in error detecting

Disadvantages:
- × DC-Components
- × complixty
- × Baseline Wandering

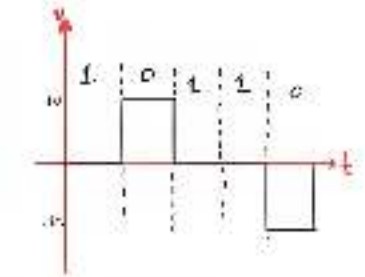
Lecture 12

Line Coding





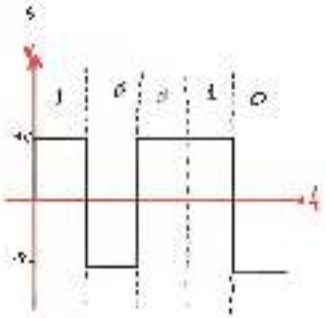
→ Alternate mark Inversion



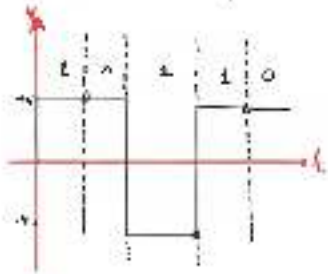
→ Pseudoternary

Polar

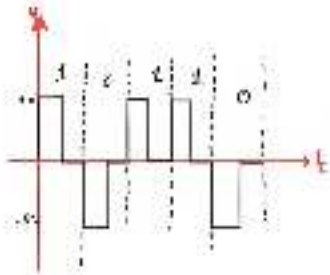
Bipolar



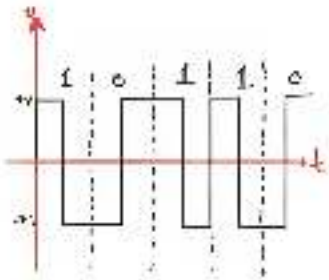
NRZ-L



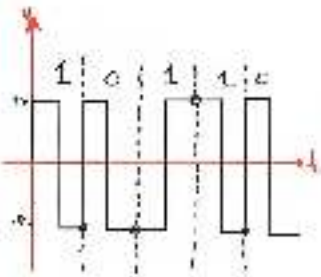
NRZ-I



RZ



Manchester



Diff. Manchester

Lecture 13

Multilevel Schemes:

mBnL

- m: number of data elements / length of data Patterns → 2^m
- B: Binary
- n: number of signal elements / length of signal Patterns
- L: number of levels (L=2 → B, L=3 → T, L=4 → Q) → L^n

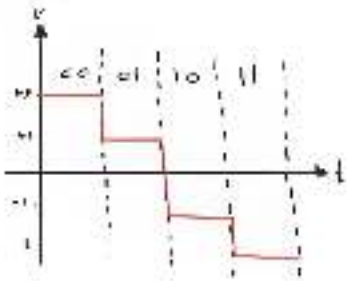
$2^m = L^n$ } → $2^m \leq L^n$

$2^m < L^n$

$2^m > L^n$ X

Ex: 2B1Q

00 → +2
01 → +1
10 → -1
11 → -3



Ex: 8B6T

2^8 3^6

$2^8 < 3^6$

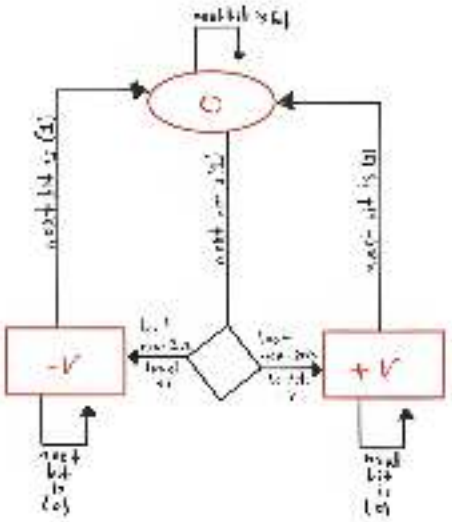
$256 < 729$

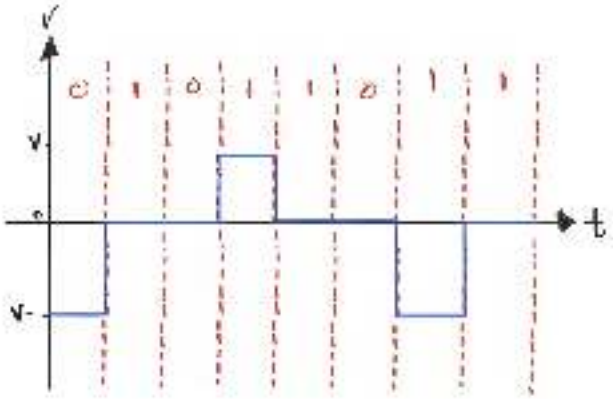
729 − 256 = 473

Redundant

Multitransmition

MLT 3L





Lecture 14

Block coding

mB/nB

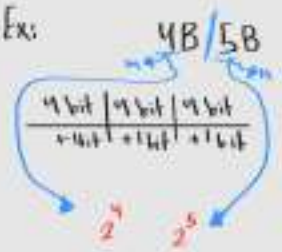
n > m

1011 0110 0110 1101

division

+ substitution → + 1bit

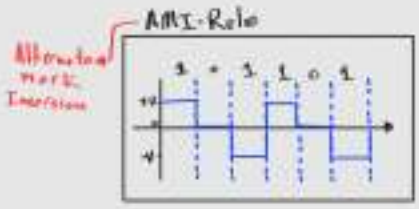
Ex: 4B/5B



$16 < 32$

$32 - 16 = 16$

for control purpose



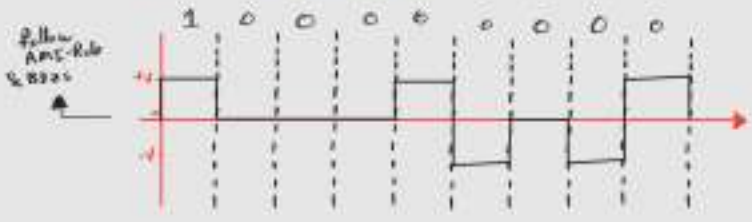
Scrambling

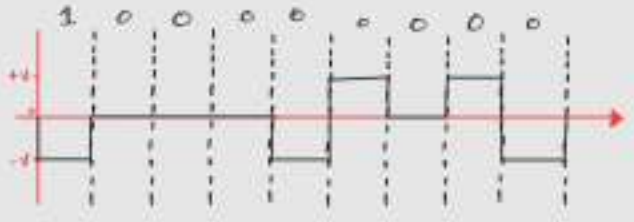
B8ZS (Bipolar with 8-zero substitution)

0000 0000

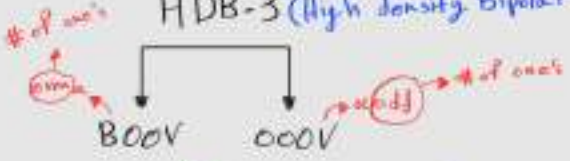
000V B0VB

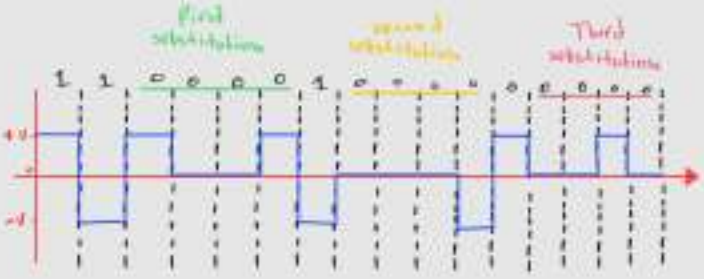
bipolar violation





HDB-3 (High density Bipolar 3-Zero)





Lecture 15

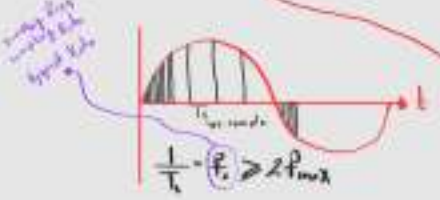
Analog signal → Analog-to-Digital Conversion → Digital signal 10101

PCM Technique
(Pulse Code Modulation)



Sampling Theorem

"Sample: To measure instantaneous value at regular time intervals"

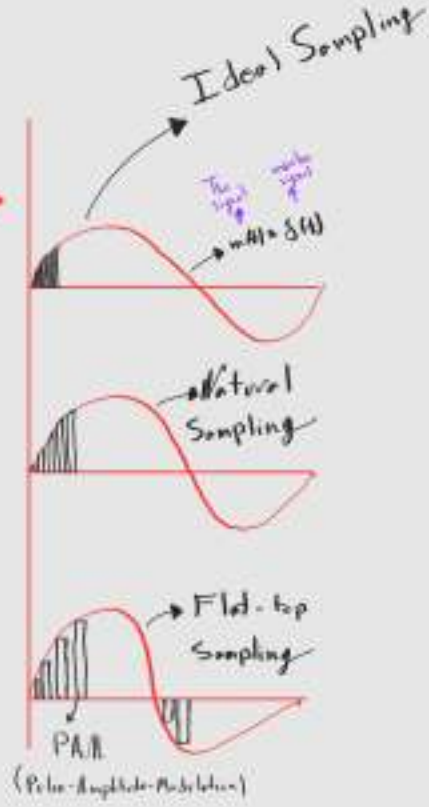


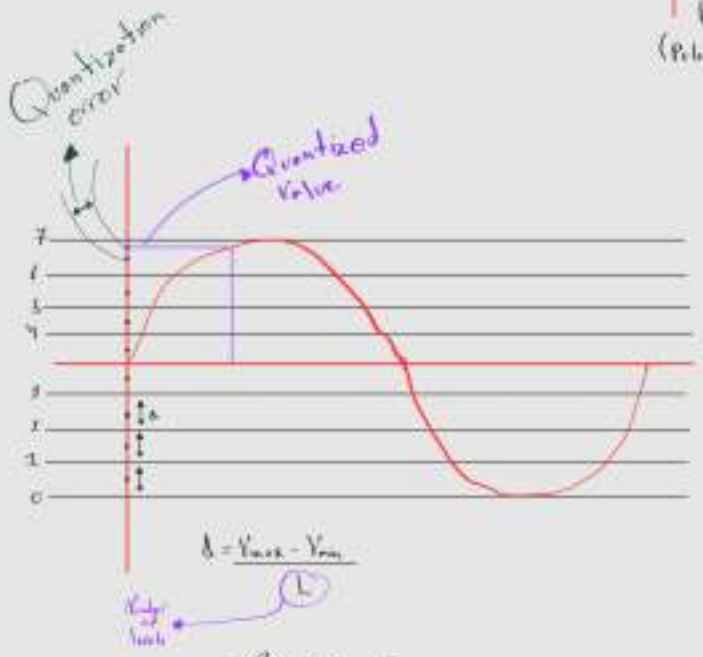
$\frac{1}{T_s} = f_s \geq 2f_{max}$

$f_s = 2f_{max}$: Sampling

$f_s >> 2f_{max}$: Over Sampling

$f_s << 2f_{max}$: Under Sampling





$\Delta = \frac{V_{max} - V_{min}}{L}$

$= \frac{20 - -20}{8} = 5$

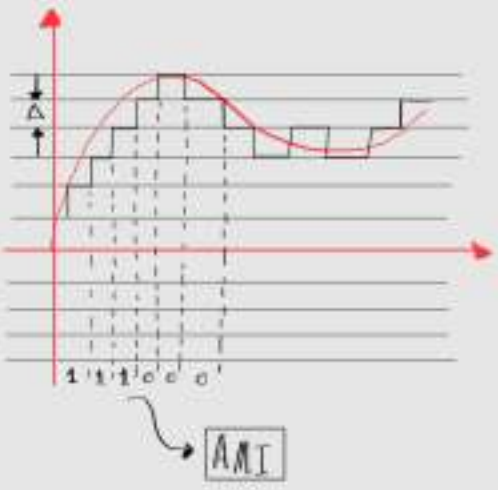
Lecture 16

Analog-to-Digital Conversion

PCM



Δ-Modulation:



$$BitRate = f_s * n_b$$

"N" — number of bits/signal level

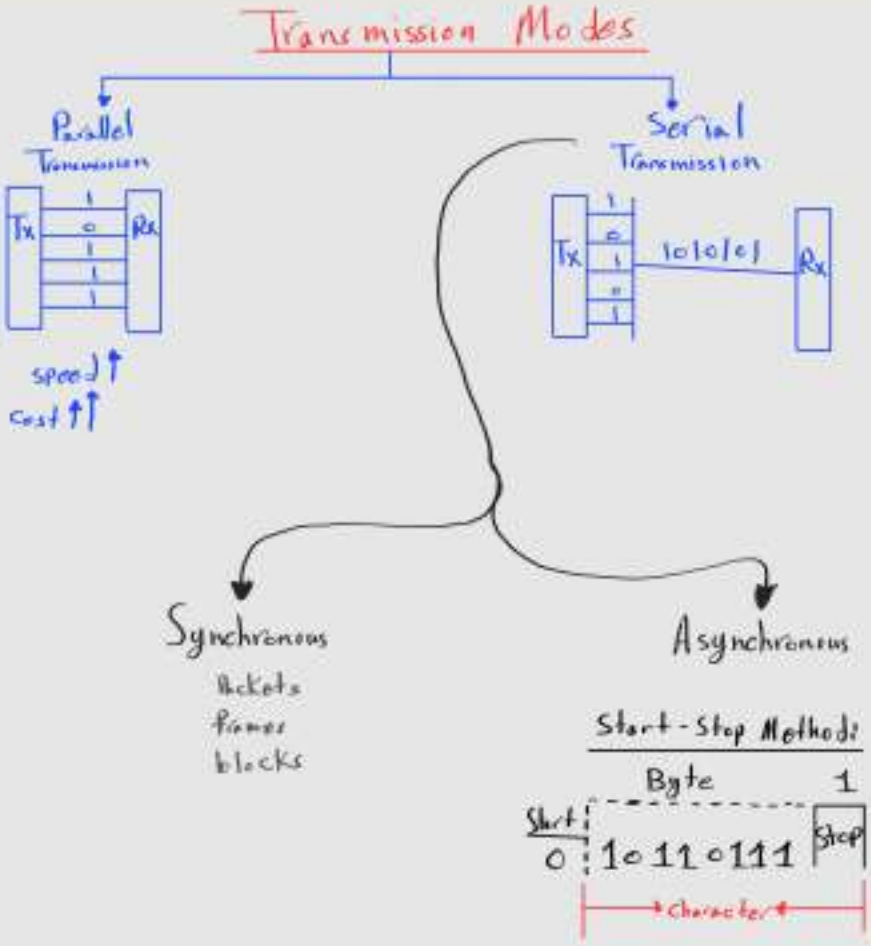
$$S_{avg} = C * \frac{N}{r}$$

$$= C * f_s * n_b * \frac{1}{r} \quad (r=1,\ C=\tfrac{1}{2})$$

$$= \frac{1}{2} * f_s * B * n_b$$

$$B_{min} = 2 * B_{analog}$$

Transmission Modes



Lecture 17

Analog Transmission

D/A conversion

Signal Rate / Baud Rate (Signal/s): $S = \frac{N}{r}$ ← Bit Rate / Data Rate (bps)

r: number of data element carried in one signal element

$r = \log_2 L$

L: number of diff. signal elements

Q: An analog signal carries 4 bits per signal element. If 1000 signal elements are sent per second. Find the bit rate

Sol:
$r = 4$
$S = 10^3$
$N = ??$

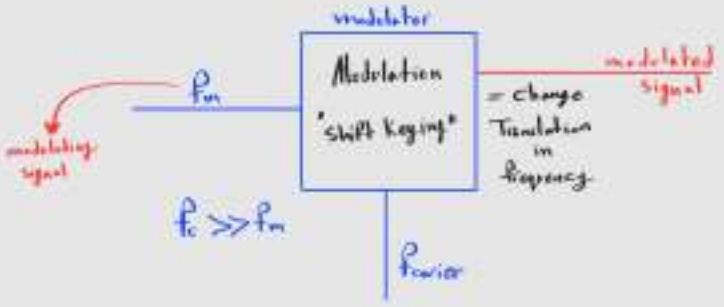
$N = Sr$
$= 4000$ bps

Q: An analog signal has a bit Rate of 8000 bps & a baud rate of 10^3 baud. How many data elements are carried by each signal element? How many signal elements do we need?

Sol:
$N = 8000$ bps
$S = 10^3$ baud
$r = ??$
$L = ??$

$\rightarrow S = \frac{N}{r}$
$r = \frac{N}{S} = \frac{8000}{1000} = 8$ bit/signal

$\rightarrow r = \log_2 L \rightarrow 8 = \log_2 L \rightarrow L = 2^8 = 256$ signal element



D/A conversion

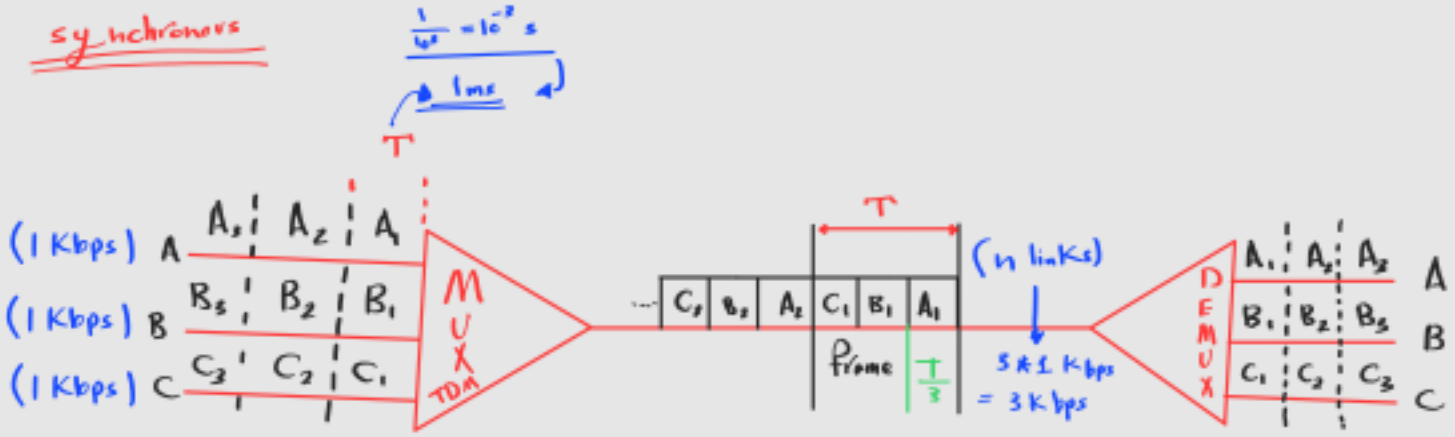
- Amplitude shift Keying "ASK"
  - on/off keying
  - $r = 1$
  - $S = N$
  - $B = (1+d)S$ ← Factor
  - Data: 1 0 1 1 0
  - Carrier
  - ASK
- Frequency shift Keying "FSK" $r = 1$
  - 1 0 1 0
  - $B = (1+d)S + 2\Delta f$
- Phase shift Keying "PSK" $r = 1$
  - $B = (1+d)S$

Lecture 18

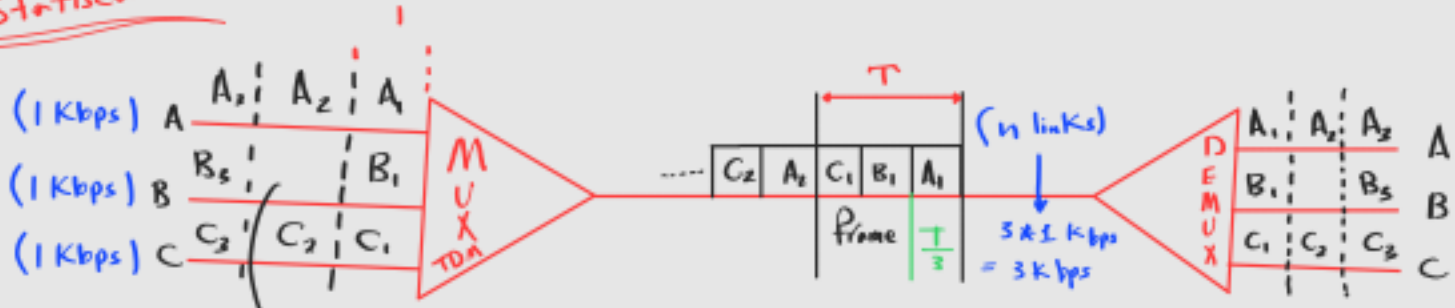
Time Division Multiplexing

-TDM-

synchronous



Statiscal



'In statiscal TDM it Improve the efficiency by removing the empty slots from the frame.

Q: Consider 3 input connections with data rate of each input is 1Kbps If 1 bit at time is multiplexed. what is the duration of:

① Each input slot.

② Each output slot.

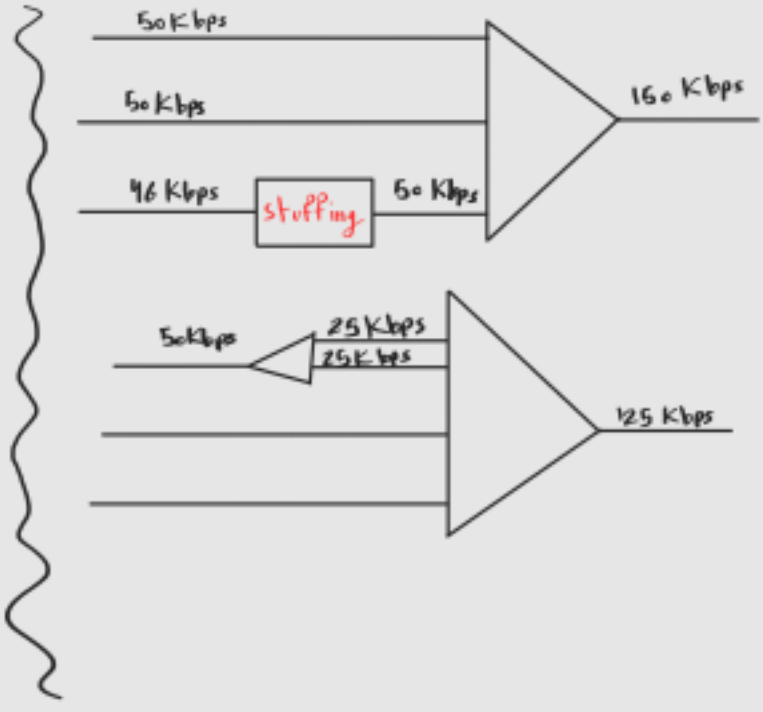
③ Each frame.

sol:

① $T = \frac{1}{1\,Kbps} = 1ms$

② $\frac{T}{3} \rightarrow \frac{1}{3}ms$

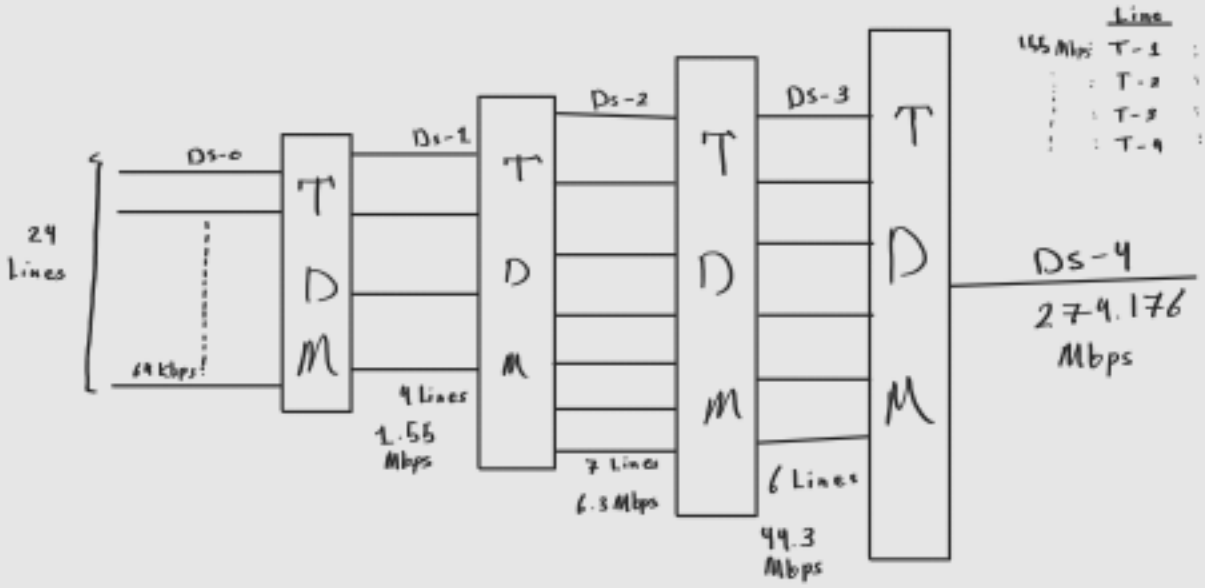
③ $T = 1ms$



Digital Signal Services

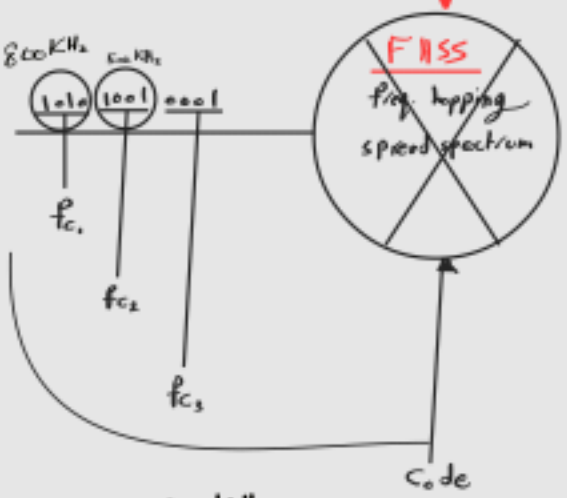
T-Lines / E-Lines

Line	service	
1.55 Mbps T-1	DS-1	24 voice channel
T-2	DS-2	96 voice channel
T-3	DS-3	
T-4	DS-4	



Lecture 19

Spread Spectrum



DSSS

Direct Sequence spread spectrum

bits → n bits

0000 → 300 KHz

0001 → 400 KHz

0010 → 500 KHz

0011 → 600 KHz

Lecture 20

Transmission Media

Sender
Tx
Source

physical Layer

receipient
Rx
destination

physical Layer

Wireless

Air = Bypass Water

Wired

Twisted pair Cable



UTP STP

Connectors: RJ-45

Coaxial Cable

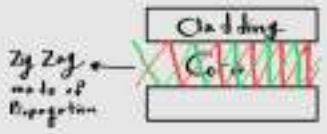


Category	Impedance	Use
RG-59	(75 Ω)	Tv
RG-58	(50 Ω)	Ethernet
RG-11	(50 Ω)	

Connectors:

- T-Connector
- Tv-Connector
- Terminator to avoid reflection

Fiber-Optic Cable



"glass" Plastic



Modes



Lecture 21

Broadcasting

Unguided (Wireless) Medivms

- air "Freespace"
- water



EM Wave

Radio Freq

EM Spectrum

RF

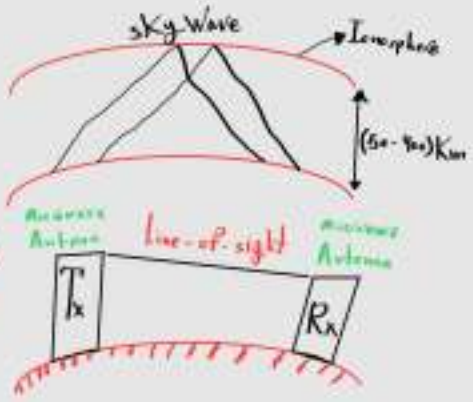
Microwave

	Band	Mode of Propagation	Applications	
Very low freq (VLF)	(3 → 30) KHz	Ground Wave	Navigation	
(LF)	(30 → 300) KHz			
(MF)	300 KHz → 3 MHz	Sky Wave	AM broadcasting	(535 → 1605) KHz
(HF)	(3 → 30) MHz		Citizen Band	
(VHF)	(30 → 300) MHz		FM radio	(88 → 108) MHz
(UHF)	300 MHz → 3 GHz	LOS	Satellite Cellular Telephony Radar	
(SHF)	(3 → 30) GHz			
(EHF)	(30 → 300) GHz			

Propagation methods:

omnidirectional Antenna

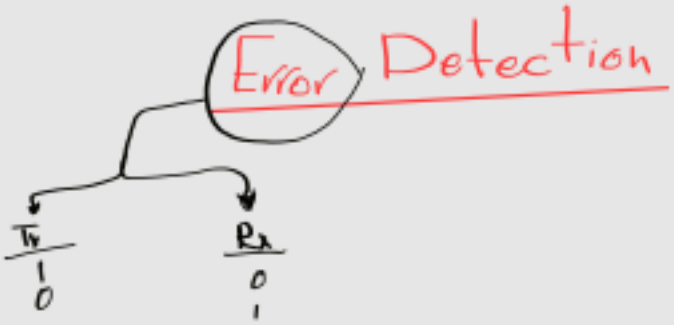


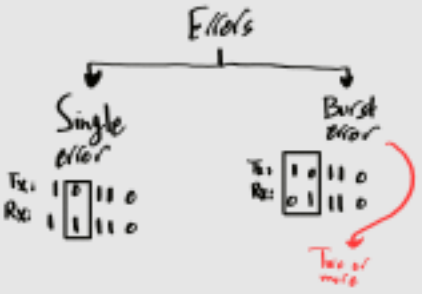


Unidirectional Antenna

Lecture 22

Error Detection

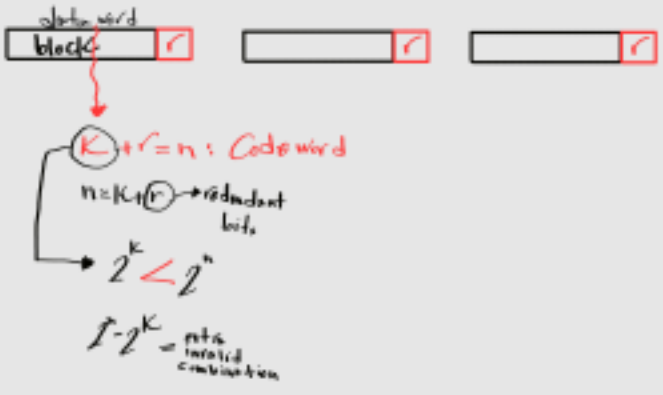




Redundancy:

To add extra bits to detect the error.

Block linear Code



Hamming Distance

$d(\overset{T_x}{1}0\overset{R_x}{1}, 111) = 1 = S$

$d(\overset{T_x}{0}1\overset{R_x}{1}, 101) = 2 = S$

$d(x \oplus y) = S$

XOR

0 0	0
0 1	1
1 0	1
1 1	0